

A Comprehensive Study on Sarcasm Detection Using Deep Learning Models

A Thesis Submitted
In Partial Fulfillment of the Requirements for the Degree of
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CANDIDATE DECLARATION

We hereby declare that the work presented in this report entitled “**A Comprehensive Study on Sarcasm Detection Using Deep Learning Models**”, submitted towards fulfillment of BACHELOR’S THESIS report in Department of Information Technology at Indian Institute of Information Technology Allahabad, is an authenticated record of our original work carried out under the guidance of **Dr. Naveen Saini**. Due acknowledgements have been made in the text to all other material used. The project was done in full compliance with the requirements and constraints of the prescribed curriculum.

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CERTIFICATE FROM SUPERVISOR

This is to certify that the statement made by the candidate is correct to the best of my knowledge and belief. The project titled **A Comprehensive Study on Sarcasm Detection Using Deep Learning Models** is a record of candidates' work carried out by them under my guidance and supervision. I do hereby recommend that it should be accepted in the fulfillment of the requirements of the Bachelor's thesis at IIIT Allahabad.

Dr. Naveen Saini

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The forgoing thesis is hereby approved as a creditable study carried out in the area of Information Technology and presented in a manner satisfactory to warrant its acceptance as a pre-requisite to the degree for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but approve the thesis only for the purpose for which it is submitted.

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Abstract

Sarcasm detection has emerged as a critical task in natural language processing, especially in the context of news headlines where subtle nuances can significantly impact interpretation. In this report, we present a meticulously curated dataset comprising of sarcastic and non-sarcastic news headlines, tailored to facilitate the development and evaluation of sarcasm detection models. Our primary objective is to explore the efficacy of various deep-learning architectures in discerning sarcasm within news headlines. We conduct experiments employing state-of-the-art models including DistilBERT, BERT, and GPT-2, SetFit each renowned for their prowess in natural language understanding tasks. This research contributes to the advancement of sarcasm detection techniques in natural language processing, offering valuable insights for researchers and practitioners alike. By shedding light on the comparative effectiveness of deep learning models in detecting sarcasm within news headlines, our study paves the way for more robust and accurate sarcasm detection systems in real-world applications.

Keywords: Sarcasm Detection, Natural Language Processing, BERT, Distill-BERT, SETFIT, GPT-2, Llama 2

Chapter 1

Introduction

Sarcasm, a form of verbal irony, has become increasingly prevalent in natural language, presenting a unique challenge for computational systems tasked with understanding human communication. In the realm of natural language processing (NLP), the detection of sarcasm holds significant importance, particularly in contexts such as news headlines where subtle nuances can drastically alter the interpretation of information. The ability to accurately discern sarcasm not only enhances the comprehension of textual content but also aids in the development of more nuanced and contextually aware NLP applications.

Recognizing the growing importance of sarcasm detection, this research endeavors to contribute to the advancement of techniques within the field of NLP by focusing specifically on the detection of sarcasm in news headlines. To facilitate this endeavor, we present a meticulously curated dataset comprising both sarcastic and non-sarcastic news headlines. This dataset serves as a crucial resource for the development and evaluation of sarcasm detection models, providing a standardized benchmark for assessing the efficacy of various approaches.

The primary objective of this study is to explore the effectiveness of deep learn-

ing architectures in discerning sarcasm within news headlines. Deep learning models have demonstrated remarkable capabilities in various NLP tasks, and we aim to leverage their potential in the domain of sarcasm detection. To this end, we employ state-of-the-art models including DistilBERT, BERT, and GPT-2, each renowned for their proficiency in natural language understanding tasks. By conducting comprehensive experiments with these models, we seek to evaluate their performance and identify strategies for improving sarcasm detection accuracy.

This research contributes to the broader landscape of sarcasm detection in NLP by providing valuable insights into the comparative effectiveness of deep learning models. By shedding light on the strengths and limitations of these models in detecting sarcasm within news headlines, our study aims to inform both researchers and practitioners in the development of more robust and accurate sarcasm detection systems. Ultimately, our findings pave the way for the integration of advanced NLP techniques into real-world applications, facilitating enhanced comprehension and interpretation of textual content in various contexts.

1.1 Motivation

The motivation behind our research stems from the increasing significance of sarcasm detection in modern NLP applications. As digital communication proliferates across social media platforms, online forums, and textual data sources, deciphering sarcasm has become paramount for understanding user sentiment, opinion mining, and social media monitoring. However, the inherent complexity

of sarcasm, intertwined with linguistic nuances and contextual cues, presents formidable challenges for automated detection systems. Thus, there is a pressing need to develop robust computational methods capable of accurately detecting sarcasm in diverse linguistic contexts and domains.

1.2 Objectives

1. **Dataset Curation:** Develop a meticulously curated dataset comprising sarcastic and non-sarcastic news headlines. This dataset will serve as a standardized benchmark for the evaluation and comparison of sarcasm detection models. Through careful selection and annotation of headlines, ensure representative coverage of various linguistic styles and contexts commonly encountered in news articles.
2. **Deep Learning Model Evaluation:** Explore the effectiveness of deep learning architectures, including DistilBERT, BERT, Llama 2 and GPT-2, in detecting sarcasm within news headlines. Conduct comprehensive experiments to assess the performance of these models, considering factors such as accuracy, precision, recall, and F1-score. Evaluate the models' ability to capture nuanced linguistic cues indicative of sarcasm and analyze their strengths and weaknesses in this context.
3. **Performance Analysis and Improvement Strategies:** Compare the performance of different deep learning models in sarcasm detection and identify strategies for enhancing accuracy. Analyze the results of the experiments

to gain insights into the factors influencing model performance, such as dataset size, training techniques, and model architecture. Based on these findings, propose recommendations for improving sarcasm detection accuracy, including potential modifications to model architectures, fine-tuning approaches, or data augmentation techniques.

1.3 Challenges

1. **Limited Availability of Sarcasm Data:** The scarcity of labeled sarcasm data, especially in niche domains, hindered the training of robust sarcasm detection models. Insufficient data could lead to overfitting or inadequate model performance, necessitating careful consideration of data augmentation and transfer learning techniques.
2. **Algorithm Selection and Tuning:** Selecting suitable deep learning algorithms and optimizing their hyperparameters posed a significant challenge. With numerous models available, determining the optimal configuration for sarcasm detection required extensive experimentation and evaluation. Balancing model complexity with computational resources and dataset size was crucial for achieving satisfactory performance.
3. **Curation of Dataset:** curation of a high-quality dataset comprising sarcastic and non-sarcastic news headlines, particularly focusing on news related to India. With only two reputable sources of sarcastic news headlines specific to India, the closure of one of these sources, "The Unreal Times," in 2016 posed a substantial obstacle. To retrieve headlines from this defunct website, reliance on archived snapshots from the Wayback Machine

became necessary. However, this introduced further complications as the Wayback Machine's API had rate limitations, impeding the efficiency of data collection and potentially impacting the research timeline

Chapter 2

Related Works

In recent years, automatic sarcasm detection has garnered significant attention within the sentiment analysis community due to its pivotal role in understanding nuanced language cues in text. Various approaches and methodologies have been explored to address this challenge. Some of the related work in Sarcasm detection are listed below:

A seminal work in the field of automatic sarcasm detection highlights the evolution of research in this domain. Initially, approaches relied on speech-based features to identify sarcasm implicitly. Subsequent advancements witnessed the adoption of semi-supervised pattern extraction techniques to uncover implicit sentiment and the integration of hashtag-based supervision for improved detection accuracy. Moreover, recent efforts have focused on incorporating contextual information beyond the target text, signifying a shift towards more comprehensive and context-aware detection methodologies [1]

The introduction of BERT revolutionized the landscape of natural language processing by offering deep bidirectional representations from unlabeled text. Unlike its predecessors, BERT leverages both left and right context in all layers,

enabling it to achieve state-of-the-art results across a wide range of NLP tasks with minimal task-specific modifications. BERT’s versatility and simplicity have propelled it to the forefront of sarcasm detection research, demonstrating remarkable performance improvements on benchmark datasets [2].

Building upon the success of BERT, RoBERTa presents a replication study that addresses key limitations in BERT’s pretraining methodology. By optimizing hyperparameters, training set size, and masking patterns, RoBERTa surpasses the performance of post-BERT methods while maintaining simplicity and efficiency. Furthermore, the introduction of a new dataset, CC-NEWS, enables better control over training set size effects, facilitating more robust and reproducible research outcomes [3].

Recognizing the computational challenges posed by large-scale pre-trained models like BERT, DistilBERT proposes a method to pre-train a smaller, general-purpose language representation model. Leveraging knowledge distillation during pre-training, DistilBERT achieves significant model compression while retaining the vast majority of language understanding capabilities. This compact and efficient model demonstrates promising results for on-device computations, offering practical solutions for resource-constrained environments [5].

Addressing the limitations of existing few-shot methods, SETFIT introduces an efficient and prompt-free framework for fine-tuning Sentence Transformers (ST) in label-scarce settings. By leveraging contrastive Siamese training and rich text embeddings, SETFIT achieves high accuracy with significantly fewer parameters compared to traditional approaches. Additionally, SETFIT demonstrates adaptability in multilingual settings, underscoring its versatility and applicability across diverse linguistic contexts [7].

In recent years, natural language processing tasks have predominantly relied on supervised learning approaches, where models are trained on task-specific datasets annotated with explicit labels. However, a groundbreaking study introduces a paradigm shift by demonstrating that language models can begin to learn these tasks without explicit supervision. This pioneering work leverages a new dataset comprising millions of webpages, termed WebText, to train large-scale language models such as GPT-2 [4].

The introduction of Llama 2 marks a significant stride in the realm of large language models (LLMs), particularly tailored for dialogue-centric tasks. Ranging from 7 billion to 70 billion parameters, Llama 2 offers pretrained and fine-tuned models optimized for dialogue scenarios, surpassing open-source chat models on various benchmarks. Noteworthy is Llama 2-Chat’s superior performance and safety evaluations, potentially making it a viable alternative to closed-source models. The project’s emphasis on fine-tuning methodologies and safety enhancements underscores its commitment to responsible LLM development. By providing detailed insights into its approach, Llama 2 facilitates community collaboration and knowledge sharing, fostering transparency and progress in the field. Overall, Llama 2 represents a pivotal advancement in LLM technology, poised to empower dialogue-based applications while prioritizing ethical considerations and community engagement [6].

Chapter 3

Dataset Creation & Preprocessing

We've created a database to tackle the shortage of good datasets containing sarcastic news headlines from India. Existing datasets mostly come from social media, which can be informal and lack context.

To fill this gap, we focused on two well-known sources of satirical news in India:

- The Fauxy¹
- The Unreal Times²



Figure 3.1: Word cloud for non-sarcastic news headlines

¹<https://thefauxy.com/>

²<https://web.archive.org/web/20240000000000/http://www.theunrealtimes.com/>

We have made the dataset publicly available³ to be used by other researchers.

Text	Sarcastic/Non-Sarcastic
Diesel costlier by 3 as Bihar government withdraws VAT rebate.	Non-sarcastic
Yogendra Yadav Beaten Up At Immigration For Applying ‘Protester Visa’ To Travel To Canada.	Sarcastic
Salman Khan’s lawyer: No evidence produced that actor drove the car	Non-sarcastic
Arvind Kejriwal To Inaugurate Covid 3rd Wave.	Sarcastic

Table 3.1: Examples of sarcastic and non-sarcastic text

3.1 Data Preprocessing

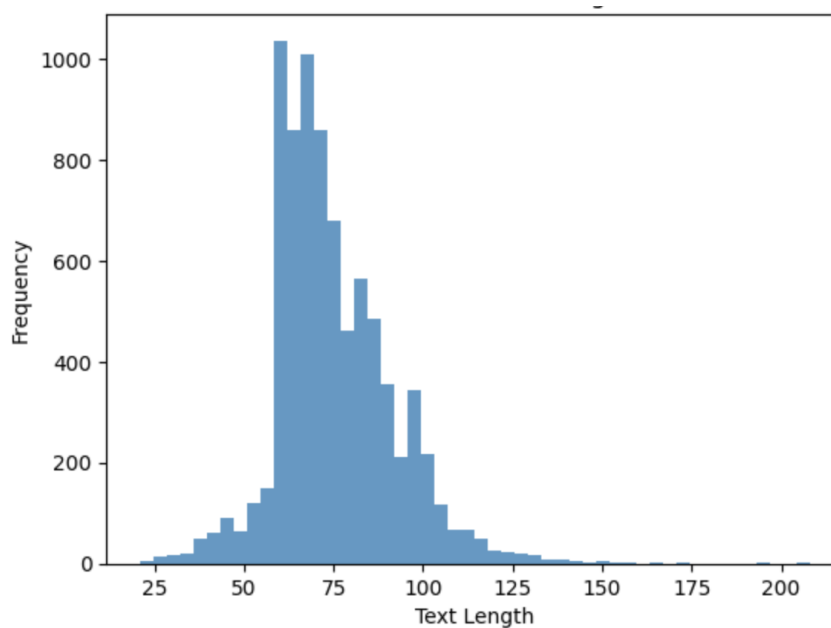


Figure 3.3: Text Length Distribution

The preprocessing pipeline for the raw extracted text data, as seen in Figure 3, entails a systematic sequence of steps to refine the dataset for subsequent

³<https://www.kaggle.com/datasets/raviutsavk/indian-news-sarcastic-headline-dataset>

analysis and modeling purposes. Initially, irrelevant columns, beyond the text content and corresponding labels, are removed to streamline the dataset.

Following this, duplicate entries are expunged to ensure data integrity and prevent skewed model training outcomes. Subsequently, textual artifacts such as HTML tags, hyperlinks, and user mentions are eliminated to mitigate noise and enhance the quality of the text corpus. Expanding contractions serves to normalize the language by converting abbreviated forms to their full lexical equivalents.

Additionally, the removal of stop words diminishes the dataset’s dimensionality by excising ubiquitous yet semantically inconsequential terms. Tokenization is then applied to segment the text into discrete units, facilitating subsequent analysis. Lastly, lemmatization is employed to standardize word forms, reducing vocabulary variation and optimizing the dataset for downstream tasks. This comprehensive preprocessing regimen ensures that the text data is suitably refined and prepared for rigorous academic analysis and modeling endeavors.

Label	Count
Sarcastic	4047
Non Sarcastic	4047

Table 3.2: Dataset Statistics

3.2 Testing and Training Split

Following preprocessing, the data underwent a standard training and testing split, with 80% allocated for training and 20% for testing. This resulted in 6438 instances for training and 1609 for testing. The training set was used to train the model parameters, while the testing set served to evaluate the model’s

performance on unseen data. This partitioning ensures a fair assessment of the model's generalization ability and reliability for real-world applications.

Chapter 4

Methodology and Model Architecture

Model architecture for our approach can be seen in Figure 4.1.

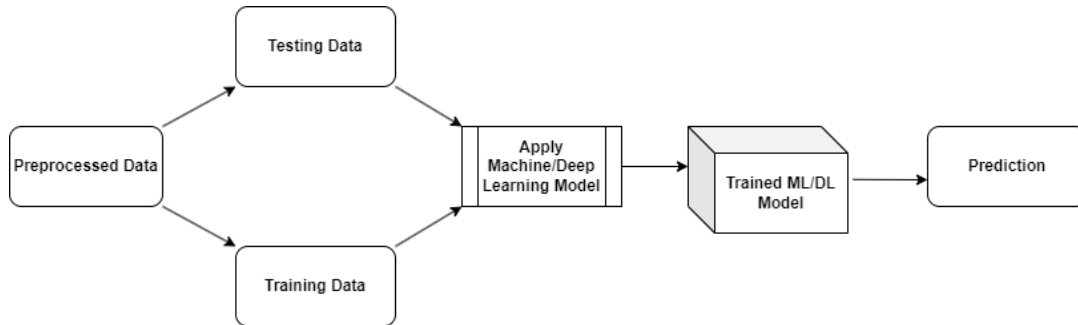


Figure 4.1: Model Architecture

4.1 Model Selection

4.1.1 DistilBERT with Classical ML Algorithms:

This model utilizes DistilBERT [5] for word embedding, capturing contextual information effectively, followed by applying classical machine learning algorithms like logistic regression, support vector machine, random forest, and naive Bayes for classification. DistilBERT extracts contextual embeddings from text,

which are then fed into classical ML algorithms to learn patterns in the data. This combination allows the model to understand nuanced language context and patterns relevant to sarcasm in news headlines.

4.1.2 DistilBERT with Frozen Parameters:

In this approach, DistilBERT is combined with a classification head while keeping the parameters of the DistilBERT model frozen. Only the classification head is trained on task-specific data. By keeping the DistilBERT parameters frozen, the model retains the pre-trained knowledge of language representations, ensuring computational efficiency. This approach focuses on training the classification layer to effectively distinguish between sarcastic and non-sarcastic headlines.

4.1.3 DistilBERT Fine-Tuning:

Similar to the previous model, this approach involves using DistilBERT with a classification head, but here, both the parameters of the DistilBERT model and the classification head are fine-tuned on the sarcasm detection task. Fine-tuning allows the model to adapt its representations specifically for sarcasm detection, potentially leading to improved performance compared to models with frozen parameters. This fine-tuning process optimizes the model's ability to capture sarcasm-related linguistic nuances and contextual cues.

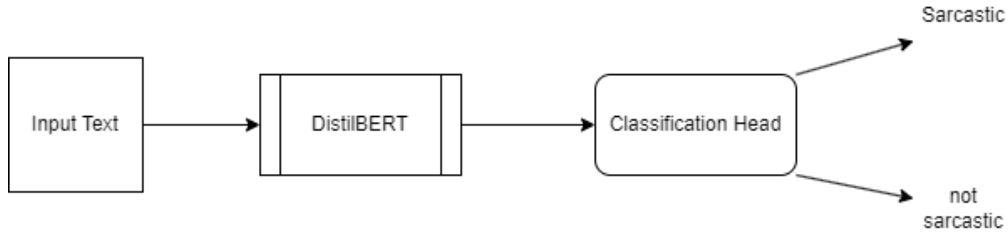


Figure 4.2: DistilBERT with classification head’s architecture

4.1.4 RoBERTa:

RoBERTa [3] is built upon the BERT architecture with additional training data and techniques. It excels in capturing intricate linguistic patterns and semantics in text data. RoBERTa undergoes extensive pre-training on a large corpus of text data, enabling it to understand complex language structures and nuances. By leveraging RoBERTa’s comprehensive contextual understanding, this model can effectively discern between sarcastic and non-sarcastic content in news headlines, even in subtle contexts.

4.1.5 SetFit:

SetFit [7] focuses on enhancing model efficiency and scalability by employing advanced transformer-based architectures. It efficiently captures complex linguistic patterns, making it suitable for processing large-scale datasets. SetFit’s efficient processing capabilities enable it to effectively capture sarcasm-related linguistic nuances and contextual cues in news headlines, making it a viable option for sarcasm detection tasks.

4.1.6 GPT2:

GPT2 [4] excels in generating coherent text and understanding contextual information due to its large-scale pre-training. With its expansive understanding of language, GPT2 is well-suited for sarcasm detection tasks in news headlines, where understanding contextual cues is crucial. GPT2's sophisticated language modeling capabilities enable it to effectively discern between sarcastic and non-sarcastic content in news headlines, making it a valuable tool for sarcasm detection tasks.

4.1.7 Llama 2:

As a state-of-the-art transformer-based model designed for natural language understanding tasks, Llama 2 [6] is adept at capturing fine-grained linguistic features and contextual information. With its advanced modeling capabilities, Llama 2 can effectively discern between sarcastic and non-sarcastic content in news headlines, leveraging its comprehensive understanding of language and contextual cues. This makes Llama 2 a potent tool for sarcasm detection tasks.

4.2 Model Training and evaluation metrics

4.2.1 DistilBERT as word embedding:

First, the DistilBERT model is initialized, along with its corresponding tokenizer and pre-trained weights. The input text data is tokenized and padded to ensure uniform length across sequences. Attention masks are created to indicate which tokens are actual words and which are padding. These inputs are

then fed into the DistilBERT model to obtain contextual embeddings. The resulting embeddings are used as features for training classical machine learning algorithms such as logistic regression, random forest, naive Bayes, and support vector machine (SVM). Each classifier is trained on the training data and evaluated on the testing data using accuracy, classification report, ROC AUC score, F1 score, and confusion matrix metrics to assess performance.

4.2.2 DistilBERT with Frozen Parameters:

The DistilBERT preprocessor and classifier are configured with a sequence length of 512, pre-trained DistilBERT models for English text, and a binary classification task. The classifier is compiled with a sparse categorical cross-entropy loss function, the Adam optimizer with a learning rate of 1e-4, and sparse categorical accuracy as the evaluation metric. The backbone of the classifier, corresponding to DistilBERT, remains non-trainable to preserve pre-trained knowledge. After splitting the dataset into training, validation, and testing sets, the model is trained for 12 epochs with a batch size of 64. Performance evaluation includes generating confusion matrices and computing the F1 score.

4.2.3 DistilBERT Fine-Tuning:

For this model, the tokenizer, derived from a pre-trained BERT model, is used to tokenize the text data, truncating sequences to a maximum length of 32 tokens and padding shorter sequences. The pre-trained DistilBERT model for sequence classification is loaded with two output labels (sarcasm and non-

sarcasm). Training arguments are defined, including output directory, logging strategy, number of training epochs, batch size, learning rate, and evaluation strategy. A custom function is implemented to compute evaluation metrics, such as accuracy, using the Hugging Face evaluate library. We initialise the training parameters for the model along with, training arguments, datasets, and callback. The model is then trained using the specified parameters with early stopping implemented to prevent overfitting. Through this process, the pre-trained DistilBERT model is fine-tuned to effectively classify sarcasm in news headlines.

4.2.4 RoBERTa:

RoBERTa, an advanced variant of the BERT model, is utilized for sarcasm detection in news headlines. It enhances BERT's architecture by refining the pre-training process, removing the next-sentence prediction (NSP) task, and employing larger batch sizes and training epochs. This adaptation fosters a deeper contextual understanding and improves downstream task performance, such as sarcasm detection.

In this implementation, the original dataset containing text and corresponding labels is loaded and split into training, validation, and test sets. These sets are saved into separate CSV files for further processing. The datasets are then loaded using the datasets library, specifying the CSV data files' paths. Tokenization is performed using the RoBERTa tokenizer, which applies padding and truncation to ensure all sequences have the same length of 256 tokens. The tokenization process transforms the text data into numerical tokens, making it suitable for model training.

The tokenized data is then fine-tuned on the sarcasm detection task, allowing RoBERTa to capture contextual information and linguistic nuances effectively. Through this process, the model learns to discern between sarcastic and non-sarcastic content in news headlines, enhancing the accuracy of sarcasm detection systems.

4.2.5 SetFit:

A SetFit model is loaded from the Hub, specifically the "sentence-transformers/all-mpnet-base-v2" model. A SetFitTrainer is created with the specified model, training and evaluation datasets, loss function (CosineSimilarityLoss), metric (accuracy), batch size, number of iterations, and epochs.

The training process involves training the SetFit model using the specified parameters and evaluating its performance on the test dataset. After training, the accuracy metric is computed and printed. Through this process, SetFit learns to effectively classify sarcasm in news headlines, demonstrating its capability as a powerful tool for text classification tasks.

4.2.6 GPT2:

GPT2, developed by OpenAI, is employed here for sarcasm detection in news headlines. It's configured for sequence classification with 3 labels. The dataset is split into train and test sets, tokenized using GPT2's tokenizer, ensuring sequences with a max length of 60 tokens. During training, the model's parameters are optimized using an AdamW optimizer with a cosine annealing learning rate scheduler. Training metrics like loss and accuracy are computed

over multiple epochs. The model's performance is evaluated on the test dataset using metrics such as accuracy and classification report. Through this process, GPT2 learns to effectively classify sarcasm in news headlines, showcasing its prowess in text classification tasks.

4.2.7 Llama 2

The LLAMA 2 model architecture, designed for sarcasm detection in text, incorporates advanced techniques such as quantization and ensemble learning to optimize performance. Utilizing a pre-trained language model, LLAMA 2 fine-tunes its parameters with specialized configurations for sarcasm detection tasks. The model employs a BitsAndBytesConfig for quantization, allowing efficient computation with reduced precision. Additionally, LLAMA 2 utilizes a LoraConfig for training, implementing specific hyperparameters tailored to causal language modeling tasks. Training is conducted with gradient checkpointing and a cosine learning rate scheduler for efficient memory usage and effective learning rate adjustment. The model is trained over multiple epochs with a TrainingArguments configuration, which includes settings for batch size, learning rate, and evaluation strategy. Finally, the SFTTrainer orchestrates the training process, utilizing the configured model, tokenizer, and datasets to train and evaluate LLAMA 2 for sarcasm detection.

Chapter 5

Results & Error Analysis

5.1 Results

5.1.1 DistilBERT as word embedding:

The various evaluation metrics for machine learning models used in combination with distilBERT as word embeddings is as shows in Figure 5.1.

As we can see Support Vector Machine is giving us the best performance with an accuracy of 87.75%.

Model	Precision	F1 Score	ROC-AUC Score
Logistic Regression	0.87	0.87	87.24
Random Forest	0.81	0.81	81.32
Naive Bayes	0.86	0.86	86.36
Support Vector Machine	0.88	0.88	87.74

Table 5.1: DistitBERT as word embedding model's result

5.1.2 DistilBERT with Frozen Parameters:

When using DistilBERT with its parameters being frozen we observed an accuracy of 85.5%. The F1 score was 0.85.

The confusion matrix for our model, as shown in Figure 5.1, provides a comprehensive evaluation of its classification performance. Model correctly identified 984 instances as positive (TP) and 1099 instances as negative (TN). However, it also misclassified 116 negative instances as positive (FP) and 228 positive instances as negative (FN).

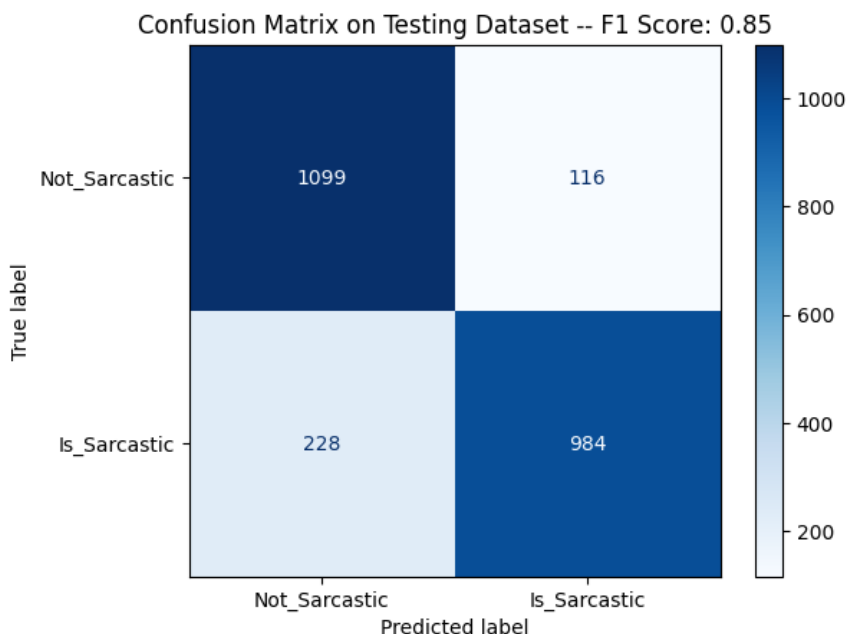


Figure 5.1: Confusion matrix of model DistilBERT with Frozen Parameters

5.1.3 DistilBERT Fine-Tuning:

An accuracy of 94.5% was observed when we finetune the distilBERT model and train it for 3 epochs. The achieved accuracy indicates that the model has effectively learned the patterns and nuances of the data, making it a reliable tool for deployment in real-world applications.

5.1.4 RoBERTa:

An accuracy of 88.7% was observed when we use RoBERTa and train it for 1 epochs.

5.1.5 SetFit:

An accuracy of 96.4% was observed with setfit. The achieved accuracy indicates that the model has effectively learned the patterns and nuances of the data, making it a reliable tool for deployment in real-world applications.

5.1.6 GPT2:

When using GPT2, we observed an accuracy of 95.0%. The F1 score was 0.95.

The confusion matrix for our model, as shown in Figure 5.2, provides a comprehensive evaluation of its classification performance. Model correctly identified 773 instances as positive (TP) and 773 instances as negative (TN). However, it also misclassified 45 negative instances as positive (FP) and 28 positive instances as negative (FN).

5.1.7 Llama 2:

When using Llama 2, we observed an accuracy of 94.8%. The F1 score was 0.95.

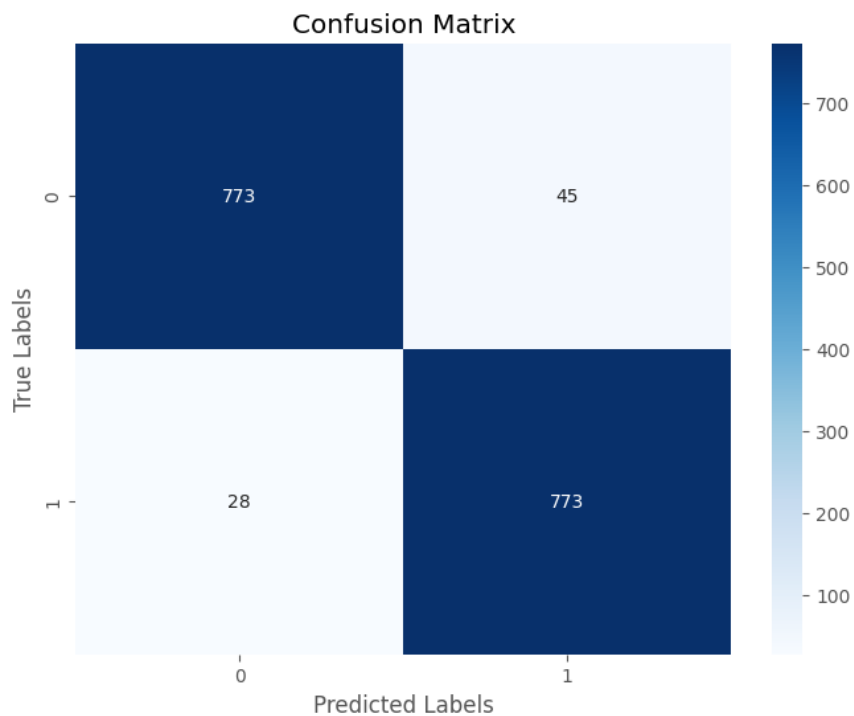


Figure 5.2: Confusion Matrix of GPT2

Model		Accuracy (in %)	F1 Score	Precision	Recall
DistilBERT as word embedding.	Logistic Regression	87.25	0.87	0.87	0.87
	Naive Bayes	81.32	0.81	0.81	0.81
	Random Forest	80.36	0.86	0.86	0.86
	Support Vector Machine	87.75	0.88	0.88	0.88
Fine tune classification head over DistilBERT		85.80	0.85	0.84	0.85
Fine tune BERT		94.50	0.94	0.94	0.95
RoBERTa		88.70	0.87	0.88	0.88
SetFit		96.4.0	0.96	0.96	0.95
GPT2		95.00	0.95	0.94	0.95
Llama 2		94.8	0.94	0.94	0.94

Table 5.2: Model's Accuracy Comparision

5.1.8 Accuracy comparison

The accuracies of the different models can be seen in the table in the Figure 5.7. As we can see, the SetFit model performs the best. Other models whose accuracies were very close to the SetFit model are distilBERT with finetuning, GPT2, Llama 2. From this observations we can conclude the models which provide us more training freedom and can be fine tuned according to our task have performed better than models with fixed parameters.

5.2 Error Analysis

In this section, we conduct a thorough examination of the errors encountered during the sarcasm detection process, shedding light on key challenges and opportunities for improvement.

- **Contextual Understanding Limitations:** A significant challenge in sarcasm detection arises from the models' difficulty in grasping contextual nuances, which are essential for accurately identifying sarcasm. Sarcasm often relies on subtle linguistic cues and context, making it challenging for models to discern sarcastic intent without a comprehensive understanding of the broader context. Misclassifications by models, where sarcastic statements are incorrectly identified, underscore the importance of improving contextual modeling techniques to enhance sarcasm detection accuracy.
- **Labeling Ambiguity:** Ensuring the accuracy of dataset labeling poses a challenge due to inherent ambiguity in sarcasm interpretation. Sarcasm can take various forms, from explicit statements to subtle, context-

dependent expressions, leading to potential mislabeling and model errors. To address this challenge, continuous refinement of labelling criteria and inclusion of diverse perspectives are necessary to minimize errors and enhance dataset quality.

Text	Correct Classification	Predicted Classification	Reason
West Indian Cricket Board approaches Mel McLaughlin to coach team.	Sarcastic	Not Sarcastic	Mel McLaughlin is a cricket news reporter, West Indies made this remark because their current coach is not delivering as expected. So they are comparing their current coach with a reporter.
SpiceJet cancels Kabul flights following spate of attacks at airport there.	Non Sarcastic	Sarcastic	There is no sarcasm in this news headline.
Taj Hotels to buy 53% of stakes of Tihar.	Sarcastic	Non Sarcastic	This headline is sarcastic because Tihar is a Jail and it doesn't make sense for a hotel to buy a jail.

Table 5.3: Error Analysis

Chapter 6

Conclusion & Future Scope

The thorough evaluation of various sarcasm detection models highlighted their distinct performance characteristics. SetFit demonstrated the highest accuracy, followed closely by fine-tuned DistilBERT, GPT2, and Llama 2. Models offering greater flexibility in training and fine-tuning parameters tended to yield superior results, emphasizing the importance of adaptability in optimizing model performance. The differences in accuracy across models underscore the significance of tailoring methodologies to task-specific requirements. SetFit's exceptional performance highlights its potential as a robust tool for sarcasm detection tasks, while insights from models like DistilBERT, GPT2, and Llama 2 contribute valuable perspectives to natural language understanding research. These findings emphasize the pivotal role of methodological diversity and model adaptability in enhancing sarcasm detection accuracy and advancing computational linguistics research. Through a comparative analysis of model accuracies, this study provides valuable insights into the efficacy of state-of-the-art models and methodologies, laying the groundwork for future research endeavors in sarcasm detection and beyond.

1. Expansion of Datasets: Incorporating larger and more diverse datasets into

sarcasm detection research can enhance model generalization and robustness. By training models on a wide range of sarcastic expressions across various contexts, researchers can improve the models' ability to detect nuanced sarcasm accurately.

2. **Integration of Multimodal Features:** Future research can explore the integration of multimodal features, such as images or user interactions, into sarcasm detection models. By incorporating additional contextual cues from different modalities, models can gain a more comprehensive understanding of sarcasm, leading to more accurate classification.
3. **Real-World Applications:** Finally, future research should focus on translating sarcasm detection advancements into practical applications across various domains, including social media analysis, customer feedback interpretation, and content moderation. By demonstrating the real-world utility of sarcasm detection technology, researchers can drive adoption and impact societal outcomes positively.

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